MA1522 Linear Algebra for Computing

Lesson 6: Properties of Determinants

Announcement: Online Quiz 1

- **Quiz 1 covers Chapter 1**, Sections 1.1–1.5: linear systems, row-echelon forms, elementary row operations, and Gaussian and Gauss–Jordan elimination.

- It opens **Saturday 29 August, 0:00** and is due **Sunday 30 August, 23:59**.

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson6` (e.g. `A1234567Z_lesson6.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 6**.
- Deadline: today, **10:30 AM**.
- If you do not know how to upload your submission, come to ask me after class.

Pollev



Basic rules:

- Please (mainly) ask the questions related to the course.

Example 1

True or false?

1. For an upper triangular matrix, the determinant is equal to the product of the diagonal entries.

2. If two rows of a square matrix are identical, then the determinant is zero.

Explain your answer to your neighbour.

(1) **True** — Section 2.8. Expand along the first column: only the $(1,1)$ entry survives, and what is left is again upper triangular, so repeating the step peels off $a_{11}a_{22}\cdots a_{nn}$.

(2) **True.** Suppose rows i and j are identical and swap them. The matrix is unchanged, so its determinant is unchanged; but a swap multiplies the determinant by -1. Hence

$$\det(A) = -\det(A), \qquad \text{so} \qquad \det(A) = 0.$$

Section 2.10 Determinant and Invertibility

Theorem

A square matrix A is invertible if and only if $\det(A) \neq 0$.

So the list of **equivalent statements for invertibility** from Section 2.6 gains one more entry. For a square matrix A of order n, the following say the same thing:

(i) A is invertible;

(v) the reduced row-echelon form of A is I_n;

(vii) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution;

(viii) for any $\mathbf{b}$, the system $A\mathbf{x} = \mathbf{b}$ has a unique solution;

(ix) $\det(A) \neq 0$.

A square matrix that is not invertible is called **singular**.

Question 1

Let $x \in \mathbb{R}$ and

$$A = \begin{pmatrix} -x & 1 & 0 \\ 0 & -x & 1 \\ 2 & -5 & 4-x \end{pmatrix}.$$

1. Compute $\det(A)$.
2. Find all x for which A is singular.
3. For each such x, find a nonzero solution of $A\mathbf{x} = \mathbf{0}$.

Question 1 — solution: parts (1) and (2)

Expand along a line with a zero in it. Row 1 will do: its entries are $-x$, 1, 0 with signs $+, -, +$, so only two 2×2 determinants survive.

$$\begin{aligned}\det(A) &= (-x)\begin{vmatrix} -x & 1 \\ -5 & 4-x \end{vmatrix} - (1)\begin{vmatrix} 0 & 1 \\ 2 & 4-x \end{vmatrix} \\ &= (-x)(x^2 - 4x + 5) - (-2) = -x^3 + 4x^2 - 5x + 2.\end{aligned}$$

Factor: $\det(A) = -(x-1)^2(x-2)$, so

$$A \text{ is singular} \iff \det(A) = 0 \iff x = 1 \text{ or } x = 2.$$

Question 1 — solution: part (3)

At each singular x the system $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution, by statement (vii). Read the first two rows downwards; the third row is then automatic.

$x = 1$: rows $1, 2$ give $b = a$ and $c = b$, so

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix},$$

row 3: $2 - 5 + 3 = 0$. ✓

$x = 2$: rows $1, 2$ give $b = 2a$ and $c = 2b$, so

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix},$$

row 3: $2 - 10 + 8 = 0$. ✓

Every scalar multiple of $\mathbf{v}_1$ is again a solution at $x = 1$, and likewise for $\mathbf{v}_2$ at $x = 2$: each singular x gives a whole line of solutions through the origin. □

Section 2.10 Determinant of a Product, a Scalar Multiple, an Inverse

Product

For square A, B of the same size,

$$\det(AB) = \det(A)\det(B),$$

and by induction
$\det(A_1 \cdots A_k) = \det(A_1) \cdots \det(A_k)$.

Scalar multiple

For A of order n and $c \in \mathbb{R}$,

$$\det(cA) = c^n \det(A).$$

There is no rule for $\det(A + B)$.

Inverse

If A is invertible, then

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

Transpose (Section 2.8)

$$\det(A^T) = \det(A).$$

Example 2

Let A, B and P be square matrices of the same size, with P invertible. True or false?

1. $\det(AB) = \det(BA)$.
2. $\det\big(PA(P^{-1})^T\big) = \det(A)$.

(1) **True.** Determinants are numbers, and numbers commute:

$$\det(AB) = \det(A)\det(B) = \det(B)\det(A) = \det(BA).$$

(2) **True.** Split the product, then transpose, then invert:

$$\det\big(PA(P^{-1})^T\big) = \det(P)\det(A)\det\big((P^{-1})^T\big) = \det(P)\det(A)\frac{1}{\det(P)} = \det(A).$$

Neither A nor B was assumed invertible here; only P was.

Section 2.10 The Adjoint

Let A be a square matrix of order n, with cofactors $A_{ij} = (-1)^{i+j} \det(M_{ij})$. The **adjoint** of A, written $\operatorname{adj}(A)$, is the $n \times n$ matrix whose (i,j)-entry is the (j,i)-cofactor of A:

$$\operatorname{adj}(A) = \begin{pmatrix} A_{11} & A_{21} & \cdots & A_{n1} \\ A_{12} & A_{22} & \cdots & A_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ A_{1n} & A_{2n} & \cdots & A_{nn} \end{pmatrix}$$

— the cofactor matrix, transposed.

Adjoint formula

$$A \operatorname{adj}(A) = \det(A)\, I_n = \operatorname{adj}(A)\, A.$$

Inverse, when $\det(A) \neq 0$

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A).$$

Question 2

Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}.$$

1. Compute the nine cofactors of A, and hence $\mathrm{adj}(A)$. You do not need to expand the nine 2×2 determinants.

2. Compute the $(1,1)$ entry of $A\,\mathrm{adj}(A)$, and explain why it is equal to $\det(A)$.

3. Compute the $(2,1)$ entry of $A\,\mathrm{adj}(A)$. Identify it with the determinant of a certain matrix, and explain why it is zero.

Question 2 — solution: part (1)

Write the cofactor matrix down with its signs and transpose it. Leave the 2×2 determinants standing — parts (2) and (3) read them, they do not need their values.

$$\operatorname{adj}(A) = \begin{pmatrix} \begin{vmatrix} 1 & 4 \\ 6 & 0 \end{vmatrix} & -\begin{vmatrix} 0 & 4 \\ 5 & 0 \end{vmatrix} & \begin{vmatrix} 0 & 1 \\ 5 & 6 \end{vmatrix} \\ -\begin{vmatrix} 2 & 3 \\ 6 & 0 \end{vmatrix} & \begin{vmatrix} 1 & 3 \\ 5 & 0 \end{vmatrix} & -\begin{vmatrix} 1 & 2 \\ 5 & 6 \end{vmatrix} \\ \begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix} & -\begin{vmatrix} 1 & 3 \\ 0 & 4 \end{vmatrix} & \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} \end{pmatrix}^T = \begin{pmatrix} \begin{vmatrix} 1 & 4 \\ 6 & 0 \end{vmatrix} & -\begin{vmatrix} 2 & 3 \\ 6 & 0 \end{vmatrix} & \begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix} \\ -\begin{vmatrix} 0 & 4 \\ 5 & 0 \end{vmatrix} & \begin{vmatrix} 1 & 3 \\ 5 & 0 \end{vmatrix} & -\begin{vmatrix} 1 & 3 \\ 0 & 4 \end{vmatrix} \\ \begin{vmatrix} 0 & 1 \\ 5 & 6 \end{vmatrix} & -\begin{vmatrix} 1 & 2 \\ 5 & 6 \end{vmatrix} & \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} \end{pmatrix}.$$

Question 2 — solution: part (2)

Row 1 of A against column 1 of $\operatorname{adj}(A)$. That column holds A_{11}, A_{12}, A_{13} — the cofactors of row 1 of A:

$$(A \operatorname{adj}(A))_{11} = 1\begin{vmatrix} 1 & 4 \\ 6 & 0 \end{vmatrix} - 2\begin{vmatrix} 0 & 4 \\ 5 & 0 \end{vmatrix} + 3\begin{vmatrix} 0 & 1 \\ 5 & 6 \end{vmatrix} = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}.$$

That last sum is exactly the cofactor expansion of $\det(A)$ along row 1. So the entry is $\det(A)$, whatever A happens to be.

Check: $-24 + 40 - 15 = 1 = \det(A)$. ✓

Question 2 — solution: part (3)

Row 2 of A against the same column — row 2's entries against row 1's cofactors:

$$(A\ \mathrm{adj}(A))_{21} = 0\begin{vmatrix}1 & 4\\ 6 & 0\end{vmatrix} - 1\begin{vmatrix}0 & 4\\ 5 & 0\end{vmatrix} + 4\begin{vmatrix}0 & 1\\ 5 & 6\end{vmatrix} = a_{21}A_{11} + a_{22}A_{12} + a_{23}A_{13}.$$

Those three cofactors are built from rows 2 and 3 of A alone. So the sum is the expansion along row 1 of the matrix B carrying row 2 of A in place of row 1, with rows 2 and 3 left alone:

$$B = \begin{pmatrix}0 & 1 & 4\\ 0 & 1 & 4\\ 5 & 6 & 0\end{pmatrix}, \qquad (A\ \mathrm{adj}(A))_{21} = \det(B).$$

B has two identical rows, so $\det(B) = 0$ by Example 1. Check: $0 + 20 - 20 = 0$. ✓ □

Question 3

Let A be an invertible square matrix of order $n \geq 2$. Show that

$$\det(\text{adj}(A)) = \det(A)^{n-1}.$$

Explain your proof to your classmates.

Question 3 — solution

Take determinants on both sides of the adjoint formula $A\,\mathrm{adj}(A) = \det(A) I_n$. The left side splits as a product; on the right, $\det(A)$ is a *scalar* multiplying I_n, so it comes out with exponent n:

$$\det(A)\det(\mathrm{adj}(A)) = \det\big(\det(A)\, I_n\big) = \det(A)^n \det(I_n) = \det(A)^n.$$

Since A is invertible, $\det(A) \neq 0$, so we may divide both sides by it:

$$\det(\mathrm{adj}(A)) = \det(A)^{n-1}.$$

□

Challenge

Question 3 assumed A invertible, and the proof needed that: it divided by $\det(A)$.

Does $\det(\text{adj}(A)) = \det(A)^{n-1}$ still hold when A is singular?

Next lesson

- **Sections 3.1–3.3 Euclidean Spaces, Dot Product, and Linear Spans**

Please watch the video clips for these sections beforehand.